

# Sophia Bian

sophiabian2029@u.northwestern.edu | +1 (650) 391-3758

## Education

### Northwestern University

Evanston, IL

*Bachelor of Science - BS & Bachelor of Music - BM*

*Expected June 2029*

Electrical Engineering & Wind Performance

Engineering Dean's Honors List

Relevant Coursework: Computer Engineering Design, Object-Oriented Programming, Design Thinking & Communication, Python, MATLAB & Linear Algebra, Electrical Engineering Fundamentals, Music Theory, Applied Flute

### Henry M. Gunn High School

Palo Alto, CA

*High School Diploma*

*June 2025*

Relevant Coursework: Principles of Engineering Design and Robotics, Digital Electronics, Physics Electricity & Magnetism, Physics Mechanics, Biology, Chemistry, Single-Variable Calculus

## Skills

Technical: Python, MATLAB, C++, Assembly Language, Computer-Aided Design, Altium Designer, NI Multisim, Circuit Design, Soldering, Machining

Soft Skills: Technical Documentation, Collaboration, Problem Solving, Project Management

Languages: English (Fluent), Chinese (Intermediate), Japanese (Elementary)

## Experience

### Northwestern Formula SAE

*Northwestern University, Evanston, IL*

Data Acquisition Subteam Member

*September 2025 - present*

- Developed features for a C++-based dashboard system to display real-time vehicle data
- Designed and tested NeoPixel drivers for visual status indicators
- Created Version 2 of a dashboard controller PCB using Altium Designer
- Collaborated in a team using Git-based workflows

### FIRST Robotics Competition Team #1868

*NASA Ames Research Center, Mountain View, CA*

Electrical Subteam Lead & Mechanical Subteam Member

*August 2022 - February 2025*

- Led electrical system architecture for a FIRST Robotics Competition team, delivering competition-ready robots on an 8-week build cycle
- Designed complete system architectures using top-down CAD modeling, specifying chassis layouts, custom mechanical subsystems, and electrical component positioning.
- Executed wiring, machining, fabrication, and full subsystem integration across mechanical and software interfaces
- Mentored team members in CAD, electronics, and manufacturing; supported two qualifications to the FIRST World Championships

## Projects

### SPICALIGN (Pediatric Medical Device Design Project)

*Northwestern University, Evanston, IL*

Member

*September 2025 - December 2025*

- Designed and prototyped an intraoperative device to position pediatric hip spica casts for car-seat compatibility
- Built a compact mechanical system using a linear actuator, servos, telescoping arms, and foot-pedal control
- Conducted user interviews with pediatric nurses and car-seat technicians to define requirements
- Collaborated with Lurie Children's Hospital on prototyping, testing, and documentation